Lesson 2 — Kohn–Sham DFT: Theory Essentials

Just enough theory to understand what the code is actually doing

Computational Materials Science for Experimentalists

Lesson at a glance (2 hours)

§	Time	Content
2.1	0:00–0:10	The many-body problem; Born–Oppenheimer; why $\rho(\boldsymbol{r})$
2.2	0:10–0:25	Hohenberg–Kohn theorems (HK1, HK2); "exact in principle"
2.3	0:25–0:55	Kohn–Sham ansatz; the SCF cycle (*deliverable: annotated diagram*)
Break	0:55–1:00	
2.4	1:00–1:25	XC functionals; Jacob's ladder; **XC error, SIE, band-gap problem, dispersion**; matching worksheet
2.5	1:25–1:45	Basis sets and pseudopotentials; **basis-set error, BSSE, PP/PAW error, frozen core**
2.6	1:45–2:00	Periodic boundary conditions, Brillouin zone, $\boldsymbol{k}$-point sampling; **$\boldsymbol{k}$-grid error, smearing, image interactions**; closing summary

Audience. Postgraduate experimentalists (spectroscopy, synthesis, characterisation) with little or no prior DFT exposure. The objective is not to derive DFT from first principles; it is to give students a working mental model of what QE and ORCA are doing internally, so they can choose inputs sensibly, recognise when results are unreliable, and report uncertainties honestly.

Pedagogical convention used in these notes. Yellow boxes are *"what to tell experimentalists"* — the bare-minimum take-home. Blue boxes are *"theory depth"* — optional extensions for motivated students or for the instructor's own reference. Red boxes flag *error sources*; green boxes give the corresponding *control strategies*. Violet boxes mark *lesson deliverables*.

Two deliverables (per the syllabus).

1. An annotated SCF cycle diagram with every term labelled (§3.4).
2. A worksheet matching XC functionals to experimental scenarios (§4.7).

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1 The Many-Body Problem and Why It Is Hard

0:00 – 0:10 (10 min)

Aim: convey *why* we cannot just solve the Schrödinger equation for a real material, and *why* $\rho(\boldsymbol{r})$ *becomes the natural variable.*

1.1 The N-electron Schrödinger equation

For N electrons in the field of M fixed nuclei the time-independent, non-relativistic Schrödinger equation reads

$$\hat{H}\,\Psi(\boldsymbol{r}_1,\ldots,\boldsymbol{r}_N) = E\,\Psi(\boldsymbol{r}_1,\ldots,\boldsymbol{r}_N), \tag{1}$$

with (in atomic units, $\hbar = m_e = e = 4\pi\epsilon_0 = 1$)

$$\hat{H} = \underbrace{-\tfrac{1}{2}\sum_{i=1}^{N}\nabla_i^2}_{\hat{T}} + \underbrace{\sum_{i=1}^{N} v_{\text{ext}}(\boldsymbol{r}_i)}_{\hat{V}_{\text{ext}}} + \underbrace{\sum_{i<j}\frac{1}{|\boldsymbol{r}_i - \boldsymbol{r}_j|}}_{\hat{V}_{ee}}, \tag{2}$$

where $v_{\text{ext}}(\boldsymbol{r}) = -\sum_{I=1}^{M} Z_I/|\boldsymbol{r} - \boldsymbol{R}_I|$ is the (electron–nucleus) external potential.

1.2 The exponential wall (Walter Kohn's term)

A direct numerical solution would require storing Ψ on a grid of N_g points in each of $3N$ spatial coordinates: storage $\sim N_g^{3N}$. For 100 electrons on a coarse 10-point grid per axis: 10^{300} values — more than the number of atoms in the observable universe. This is the **exponential wall**: the difficulty grows exponentially with system size.

What to tell experimentalists

You cannot solve the Schrödinger equation for any real molecule or crystal by brute force. *Every* electronic-structure method (HF, DFT, CCSD(T), QMC, ...) is a strategy for trading rigour against tractability. DFT trades the $3N$-dimensional wave function for the 3-dimensional density $\rho(\boldsymbol{r})$ — formally exact, practically approximate.

1.3 The Born–Oppenheimer approximation

Nuclei are $\sim 1800\times$ heavier than electrons, so on the electronic timescale the nuclei are effectively static. We therefore solve the *electronic* problem at fixed nuclear positions $\{\boldsymbol{R}_I\}$ and obtain the *electronic* energy $E_e(\{\boldsymbol{R}_I\})$. Adding the (classical) nuclear repulsion $E_{\text{nn}} = \frac{1}{2}\sum_{I\neq J} Z_I Z_J/|\boldsymbol{R}_I - \boldsymbol{R}_J|$ gives the **Born–Oppenheimer potential energy surface**

$$E_{\text{PES}}(\{\boldsymbol{R}_I\}) = E_e(\{\boldsymbol{R}_I\}) + E_{\text{nn}}(\{\boldsymbol{R}_I\}). \tag{3}$$

Everything in this course (geometry optimisation, phonons, IR/Raman spectra) is geometry on, or curvature of, this PES.

Theory depth (optional, for the curious)

BO assumes the electronic gap $|\Delta E_{e,n}|$ is large compared with vibrational quanta $\hbar\omega_{\text{vib}}$. It fails for (i) conical intersections (photochemistry), (ii) Jahn–Teller systems, (iii) some metallic / narrow-gap systems where electron-phonon coupling cannot be treated perturbatively. Outside these cases, BO is an excellent approximation and underlies essentially all of computational materials science.

1.4 Why the density $\rho(\boldsymbol{r})$?

The electron density

$$\rho(\boldsymbol{r}) = N\int |\Psi(\boldsymbol{r}, \boldsymbol{r}_2, \ldots, \boldsymbol{r}_N)|^2 \,\mathrm{d}\boldsymbol{r}_2 \cdots \mathrm{d}\boldsymbol{r}_N \tag{4}$$

is a function of *three* variables, not $3N$. It is also a directly observable quantity (X-ray diffraction). If we can recast quantum mechanics so that the ground-state energy is a functional $E[\rho]$, the exponential wall is bypassed. This is the promise of DFT — proven rigorously by the Hohenberg–Kohn theorems (§2.2).

Numerical scale of the gain: storing Ψ for 100 electrons on a 10^3 grid would require 10^{300} numbers; storing $\rho(\boldsymbol{r})$ on the same grid requires $10^3 = 1000$ numbers.

2 The Hohenberg–Kohn Theorems

0:10 – 0:25 (15 min)

Aim: state the two HK theorems, sketch the proofs at a conceptual level, and emphasise the distinction between "exact in principle" and "exact in practice".

2.1 HK-I: the density determines everything

For a non-degenerate ground state, the ground-state density $\rho_0(\boldsymbol{r})$ uniquely determines the external potential $v_{ext}(\boldsymbol{r})$ up to an additive constant, and therefore uniquely determines the Hamiltonian and every electronic observable.

Proof sketch (reductio ad absurdum). Suppose two different external potentials v_{ext} and v'_{ext} (differing by more than a constant) both produced the same ground-state density ρ_0. Their Hamiltonians $\hat{H}$ and $\hat{H}'$ differ; their ground states Ψ_0 and Ψ'_0 differ. Apply the variational principle of QM:

$$E_0 = \langle \Psi_0 | \hat{H} | \Psi_0 \rangle < \langle \Psi'_0 | \hat{H} | \Psi'_0 \rangle = E'_0 + \int (v_{\text{ext}} - v'_{\text{ext}}) \rho_0 \, \mathrm{d}\boldsymbol{r}, \tag{5}$$

$$E'_0 = \langle \Psi'_0 | \hat{H}' | \Psi'_0 \rangle < \langle \Psi_0 | \hat{H}' | \Psi_0 \rangle = E_0 + \int (v'_{\text{ext}} - v_{\text{ext}}) \rho_0 \, \mathrm{d}\boldsymbol{r}. \tag{6}$$

Adding the two strict inequalities gives $E_0 + E'_0 < E'_0 + E_0$ — a contradiction. Therefore the assumption fails: $v_{\text{ext}} \to \rho_0$ is one-to-one (up to a constant).

Logical chain.

$$\rho_0(\boldsymbol{r}) \xrightarrow{\text{HK-I}} v_{\text{ext}}(\boldsymbol{r}) \implies \hat{H} \implies \Psi_0 \implies \langle \Psi_0 | \hat{O} | \Psi_0 \rangle \text{ for any observable } \hat{O}.$$

2.2 HK-II: a variational principle for the density

There exists a universal functional $F[\rho]$ (independent of v_{ext}) such that

$$E_v[\rho] = F[\rho] + \int v_{\text{ext}}(\boldsymbol{r}) \rho(\boldsymbol{r}) \, \mathrm{d}\boldsymbol{r} \;\geq\; E_0,$$

with equality if and only if $\rho = \rho_0$.

So one could in principle find the ground state by minimising $E_v[\rho]$ over densities — a 3-D minimisation problem replacing a $3N$-D one.

2.3 Universal but unknown

$F[\rho] = T[\rho] + V_{ee}[\rho]$ is *universal*: it depends only on the electron count and the electron–electron interaction, not on the system's external potential. It is therefore the same functional for the H_2 molecule, for solid copper, for liquid water, and for a virus.

The catch: nobody knows what $F[\rho]$ is. In particular, $T[\rho]$ as an explicit functional of ρ is unknown and is the principal stumbling block.

Theory depth (optional, for the curious)

Early "orbital-free" DFT (Thomas–Fermi, 1927; Thomas–Fermi–Dirac, 1930) used

$$T_{\text{TF}}[\rho] = C_F \int \rho(\boldsymbol{r})^{5/3} \, \mathrm{d}\boldsymbol{r}, \qquad C_F = \tfrac{3}{10}(3\pi^2)^{2/3},$$

plus the Slater exchange $-C_x \int \rho^{4/3}\,\mathrm{d}\boldsymbol{r}$. *Teller's no-binding theorem* (1962): pure Thomas–Fermi theory predicts that no molecule is bound. The functional is too crude; it misses the shell structure of atoms entirely. This is what motivates Kohn–Sham.

2.4 "Exact in principle, approximate in practice"

The HK theorems prove that an *exact* energy functional of the density *exists*. They give no recipe for constructing it. Every practical DFT calculation therefore replaces the exact $F[\rho]$ with an approximation. The skill of using DFT lies in knowing which approximation is acceptable for which problem — the substance of §4–§6 of this lesson.

What to tell experimentalists

DFT is not a model; it is an exact reformulation of quantum mechanics. The approximations live entirely in the *exchange–correlation* functional we will meet in §2.4. Anything reported as a "DFT result" is really a result of "DFT + chosen XC + chosen basis + chosen pseudopotential + chosen $\boldsymbol{k}$-grid". **All five must be reported in a publication.**

3 The Kohn–Sham Ansatz and the SCF Cycle

0:25 – 0:55 (30 min)

Aim: introduce the auxiliary non-interacting system, write the KS equations, label every term in v_{eff}, and produce the annotated SCF diagram that is the lesson deliverable.

3.1 The Kohn–Sham mapping (Kohn–Sham 1965)

The interacting many-body problem is replaced by an *auxiliary* non-interacting system of N electrons moving in an effective potential $v_{\text{eff}}(\boldsymbol{r})$ chosen such that the auxiliary system has *the same ground-state density* as the real interacting one. The auxiliary electrons obey

$$\boxed{\left[-\frac{\hbar^2}{2m_e}\nabla^2 + v_{\text{eff}}(\boldsymbol{r})\right]\psi_i(\boldsymbol{r}) = \varepsilon_i\,\psi_i(\boldsymbol{r})} \tag{7}$$

and the density is

$$\rho(\boldsymbol{r}) = \sum_{i=1}^{N_{\text{occ}}} f_i\,|\psi_i(\boldsymbol{r})|^2, \tag{8}$$

with occupations $f_i \in [0,2]$ (closed shell uses $f_i = 2$ for the $N/2$ lowest orbitals; metals require partial occupancies, see §3.5.2).

The $\{\psi_i\}$ are called **Kohn–Sham orbitals**. They are mathematical constructs of the auxiliary system, *not* true single-particle wavefunctions of the interacting system.

3.2 The exchange–correlation functional defined

The total energy is decomposed as

$$\boxed{E_{\text{tot}}[\rho] = T_{\text{s}}[\rho] + \int v_{\text{ext}}(\boldsymbol{r})\rho(\boldsymbol{r})\,\mathrm{d}\boldsymbol{r} + E_{\text{H}}[\rho] + E_{\text{xc}}[\rho] + E_{\text{nn}}} \tag{9}$$

where, term by term:

- $T_{\text{s}}[\rho] = \sum_i f_i \langle\psi_i| -\frac{1}{2}\nabla^2|\psi_i\rangle$ — kinetic energy of the *non-interacting* reference (computed exactly from the KS orbitals, not from ρ directly);

- $\int v_{\text{ext}}\rho\,\mathrm{d}\boldsymbol{r}$ — electron–nucleus attraction;
- $E_{\text{H}}[\rho] = \frac{1}{2}\iint \frac{\rho(\boldsymbol{r})\rho(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}\boldsymbol{r}\,\mathrm{d}\boldsymbol{r}'$ — classical (Hartree) Coulomb;
- $E_{\text{xc}}[\rho]$ — *everything else*: exchange, correlation, and the kinetic difference $T[\rho] - T_{\text{s}}[\rho]$;
- E_{nn} — classical nucleus–nucleus repulsion.

Crucial conceptual point: the first four terms (and the fifth) are either exactly computable from ρ or exactly computable from the KS orbitals. *The entire approximation in a KS-DFT calculation lives in $E_{xc}[\rho]$.* Choose this functional, and you have specified a DFT method.

3.3 The effective potential

Functional differentiation of (9) with the constraint $\int \rho\,\mathrm{d}\boldsymbol{r} = N$ gives

$$\boxed{v_{\text{eff}}(\boldsymbol{r}) = v_{\text{ext}}(\boldsymbol{r}) + v_{\text{H}}(\boldsymbol{r}) + v_{\text{xc}}(\boldsymbol{r})} \tag{10}$$

with

$$v_{\text{ext}}(\boldsymbol{r}) = -\sum_{I=1}^{M}\frac{Z_I}{|\boldsymbol{r}-\boldsymbol{R}_I|} \quad \text{(electron–nucleus attraction; fixed once geometry is set),} \tag{11}$$

$$v_{\text{H}}(\boldsymbol{r}) = \int \frac{\rho(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}\boldsymbol{r}' \quad \text{(classical electron–electron repulsion; the \textit{Hartree potential}),} \tag{12}$$

$$v_{\text{xc}}(\boldsymbol{r}) = \frac{\delta E_{\text{xc}}[\rho]}{\delta\rho(\boldsymbol{r})} \quad \text{(everything else; \textit{the} approximate ingredient).} \tag{13}$$

The KS equations (7) are formally a one-particle Schrödinger equation, but v_{eff} depends on ρ, which depends on $\{\psi_i\}$, which solve the equation. This is a fixed-point problem: solved iteratively by the **self-consistent field (SCF) cycle**.

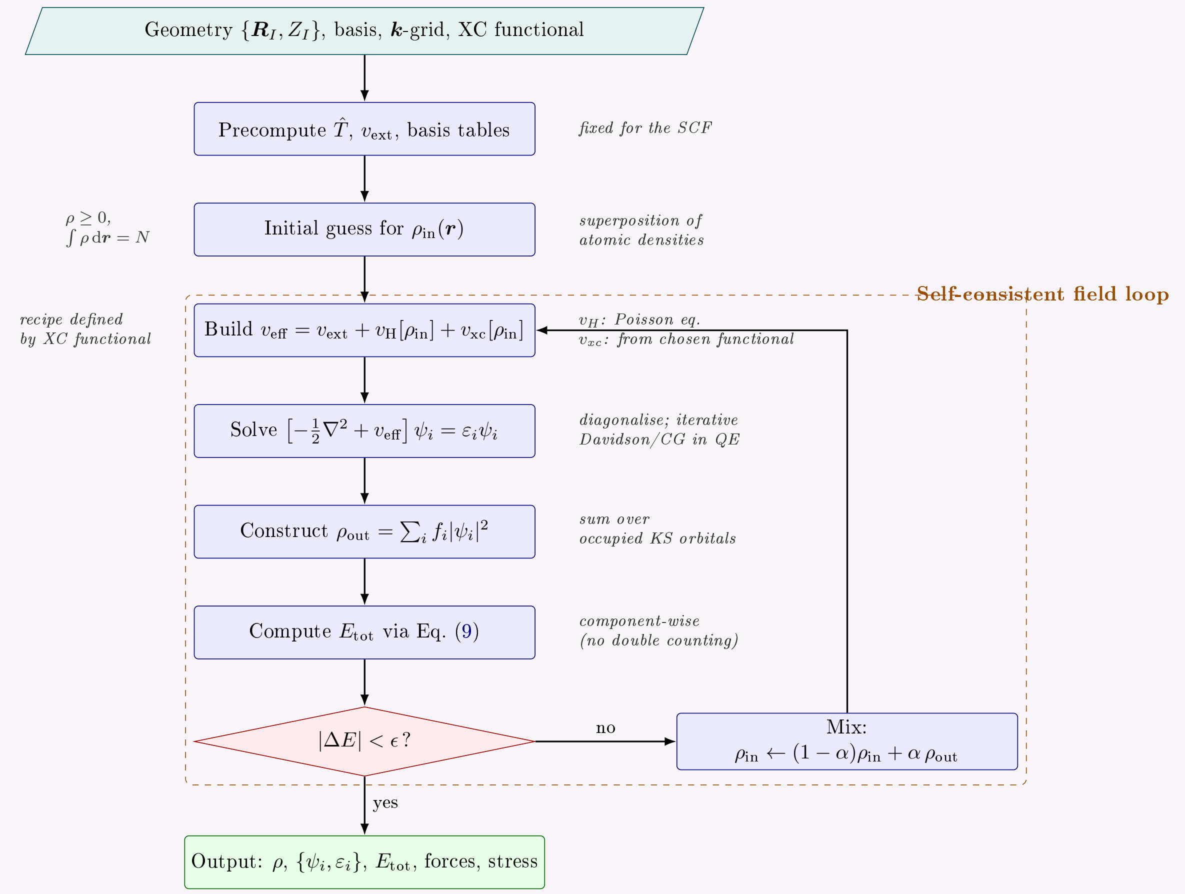
What to tell experimentalists

The KS equations look like one-electron Schrödinger equations. *The reason DFT can be so cheap is precisely that it replaces an N-electron problem with N* single-*electron problems coupled only through the average density.* The price for this enormous simplification is that we do not know v_{xc} exactly — and that is what XC functionals approximate.

3.4 The SCF cycle, step by step

This is the central deliverable of the lesson. The diagram below labels every term and every decision; it should reproduce in a single picture the entire content of (7)–(10).

Deliverable: annotated SCF cycle



Key choices the user controls.

Choice	QE keyword	ORCA keyword
XC functional	`input_dft` (or implicit in PP)	first input line, e.g. `! PBE`
Basis / cutoff	`ecutwfc`, `ecutrho`	e.g. `! def2-TZVP`
Pseudopotential / ECP	chosen `.UPF` file	`! def2/J`, `ECP`
$\boldsymbol{k}$-grid	`K_POINTS` card	— (molecules: not needed)
SCF tolerance	`conv_thr`	`TIGHTSCF` / `VERYTIGHTSCF`
Mixing	`mixing_beta`, `mixing_mode`	DIIS managed internally
Smearing (metals)	`occupations`, `degauss`	molecules: typically not used

3.5 SCF convergence in practice: numerical pitfalls

The SCF problem is a non-linear fixed-point iteration. Three things can go wrong:

Error source: SCF non-convergence (charge sloshing, oscillation)

Linear mixing $\rho_{\text{in}} \leftarrow (1-\alpha)\rho_{\text{in}} + \alpha\rho_{\text{out}}$ with too large α oscillates; too small α converges slowly. Metals are particularly prone to "charge sloshing" (long-wavelength density oscillations between iterations).

Controlling the error

(a) Pulay (DIIS) mixing: extrapolates the new density as a least-squares combination of previous iterates, minimising the commutator $[\mathbf{F}, \mathbf{P}]\mathbf{S}$. Standard in ORCA and the default in modern QE (`mixing_mode = 'plain'` or `'local-TF'` for metals). **(b) Kerker precondition-**

ing (`mixing_mode = 'TF'`, `mixing_mode = 'local-TF'` in QE) suppresses long-wavelength components — essential for metals. **(c) Level shifting / damping** pushes virtuals up to enlarge the HOMO–LUMO gap and reduce iteration-to-iteration orbital reshuffling. **(d) Better initial guess**: superposition of atomic densities (QE default), extended Hückel, or a converged density from a cheaper functional.

3.5.1 Convergence criteria

Two thresholds matter:

- **Energy change** $|E_{\text{tot}}^{(n)} - E_{\text{tot}}^{(n-1)}| < \epsilon_E$ — the physically meaningful criterion. QE: `conv_thr` (default 10^{-6} Ry). ORCA: `TIGHTSCF` (10^{-8} Ha) or `VERYTIGHTSCF` for vibrational frequencies.
- **Density change** $\|\rho^{(n)} - \rho^{(n-1)}\|$ — complementary; can mislead in non-orthogonal local bases where the density-matrix norm is dominated by the metric, not the electron count.

What to tell experimentalists

Rule of thumb for the SCF threshold. For total energies and geometries, `conv_thr = 1e-8 Ry` ($\sim 10^{-9}$ Ha) is usually safe. For *phonons* (§Lesson 4) the tolerance must be tighter — typically `conv_thr = 1e-12 Ry` — because phonon frequencies are second derivatives and small SCF noise is hugely amplified.

3.5.2 Smearing for metals

In a metal the highest occupied bands cross the Fermi level. Strict zero-temperature occupations $f_i \in \{0, 2\}$ cause discontinuous jumps in occupation as the SCF density changes between iterations, preventing convergence. The cure is a small electronic "temperature"

$$f_i = \mathcal{F}\left(\frac{\varepsilon_i - \varepsilon_F}{\sigma}\right) \tag{14}$$

with $\sigma \sim 0.005$–0.02 Ry. Choices for $\mathcal{F}$ available in QE: Fermi–Dirac (`fermi-dirac`); Gaussian (`gaussian`); Methfessel–Paxton (`methfessel-paxton`, $\boldsymbol{k}$-convergence improved); **Marzari–Vanderbilt cold smearing** (`cold`, recommended default for metals — entropy contribution is small even at finite σ).

Error source: Smearing artefact

Total energies obtained with finite σ contain a fictitious entropy $-\sigma S$. Forces, stresses, and reaction energies for metallic systems all depend on σ.

Controlling the error

Converge total energies w.r.t. σ (e.g. $\sigma = 0.02, 0.01, 0.005$ Ry) at fixed $\boldsymbol{k}$-grid; *simultaneously* converge w.r.t. $\boldsymbol{k}$-grid at fixed σ (a finer grid permits smaller σ). Marzari–Vanderbilt cold smearing has the mildest residual artefact.

3.6 Spin treatment

Closed-shell systems: *restricted* KS (RKS), α and β orbitals identical, single density. Open-shell systems: *unrestricted* KS (UKS, two densities $\rho_\uparrow, \rho_\downarrow$) or *restricted open-shell* (ROKS).

Error source: Spin contamination in UKS

A broken-symmetry UKS solution generally has $\langle \hat{S}^2 \rangle > S(S+1)$. For radicals and transition-metal complexes this contamination can shift relative energies by several kcal/mol.

Controlling the error

ORCA prints $\langle \hat{S}^2 \rangle$ in every UKS output; a deviation > 0.1 from the ideal value warrants attention. Spin-projection methods (Yamaguchi, approximate spin projection) and ROKS are partial remedies; multi-reference methods (CASSCF/NEVPT2) are the rigorous fix.

Break: 0:55 – 1:00 (5 min)

4 Exchange–Correlation Functionals — a Practical Guide

1:00 – 1:25 (25 min)

Aim: climb Jacob's ladder; expose the structural errors of approximate E_{xc} (self-interaction, delocalisation, band-gap underestimation, dispersion, static correlation); finish with the XC–experiment matching worksheet.

4.1 Where the modelling error lives

Recall: in (9) every term except $E_{\text{xc}}[\rho]$ is computable to arbitrary numerical accuracy given ρ and the orbitals. **Every theoretical error in KS-DFT is therefore an error in $E_{\text{xc}}[\rho]$.** (Basis, $\boldsymbol{k}$-grid, pseudopotential errors are separate, and discussed in §5–§6.)

4.2 Jacob's ladder of XC functionals

Perdew's pedagogical hierarchy: each rung uses richer ingredients than the previous; accuracy and cost both increase.

Rung	Class	Ingredients	Examples	Cost
1	LDA	ρ	PZ81, VWN, PW92	$\mathcal{O}(N^3)$
2	GGA	$\rho, \lvert\nabla\rho\rvert$	PBE, BLYP, PW91, PBEsol	$\mathcal{O}(N^3)$
3	meta-GGA	$+\,\tau$ (and/or $\nabla^2\rho$)	SCAN, r^{2}SCAN, TPSS, M06-L	$\mathcal{O}(N^3)$
4	hybrid	+ fraction of exact (HF) exchange	B3LYP, PBE0, HSE06	$\mathcal{O}(N^4)$
4′	range-sep. hybrid	SR/LR Coulomb split	CAM-B3LYP, ωB97X-V, LC-ωPBE	$\mathcal{O}(N^4)$
5	double hybrid	+ virtual-orbital (MP2-like) corr.	B2PLYP, DSD-PBEP86	$\mathcal{O}(N^5)$

LDA: $E_{\text{xc}}^{\text{LDA}}[\rho] = \int \rho(\boldsymbol{r})\,\varepsilon_{\text{xc}}^{\text{unif}}(\rho(\boldsymbol{r}))\,\mathrm{d}\boldsymbol{r}$. Exchange: Slater, $\varepsilon_x^{\text{unif}} = -\frac{3}{4}(3/\pi)^{1/3}\rho^{1/3}$. Correlation: VWN or PW92 (parameterisation of QMC data for the uniform electron gas). Surprisingly good for densely packed solids (the local density is locally near-uniform). *Overbinds* (~ 1 eV per bond); bond lengths too short by ~ 0.02 Å; lattice constants $\sim 1\%$ small.

GGA: adds $\lvert\nabla\rho\rvert$. PBE (non-empirical, satisfies many exact constraints) is the workhorse of solid-state physics. *Underbinds* relative to LDA; lattice constants $\sim 1\%$ large for many materials; barrier heights still $\sim$ 3–5 kcal/mol low. PBEsol re-tunes for densely-packed solids and surfaces.

meta-GGA: adds the KS kinetic-energy density $\tau(\boldsymbol{r}) = \frac{1}{2}\sum_i f_i|\boldsymbol{\nabla}\psi_i(\boldsymbol{r})|^2$ to distinguish covalent / metallic / weak-bond regions. SCAN (Sun–Ruzsinszky–Perdew, 2015) satisfies all 17 known exact constraints of a meta-GGA; r^{2}SCAN improves its numerical stability. Typical gain: thermochemistry, geometries, weak interactions all improved over PBE at $\sim$ 20–30% extra cost.

Hybrid functionals: mix exact (HF) exchange,

$$E_{\mathrm{xc}}^{\mathrm{hybrid}} = a\,E_{\mathrm{x}}^{\mathrm{HF}} + (1-a)\,E_{\mathrm{x}}^{\mathrm{DFA}} + E_{\mathrm{c}}^{\mathrm{DFA}}, \tag{15}$$

which reduces self-interaction error (§4.3) and improves band gaps, barrier heights, and thermochemistry. B3LYP ($a \approx 0.20$, three empirical parameters; the dominant choice in molecular quantum chemistry). PBE0 ($a = 1/4$, no fit). HSE06 (Heyd–Scuseria–Ernzerhof): range-separated — exact exchange only at short range, PBE at long range — the standard for solid-state band gaps (extended systems make full HF exchange prohibitive because of the long range of $1/r$).

Range-separated hybrids: split the Coulomb interaction,

$$\frac{1}{r_{12}} = \frac{\mathrm{erfc}(\omega r_{12})}{r_{12}} + \frac{\mathrm{erf}(\omega r_{12})}{r_{12}}, \tag{16}$$

with the short-range piece handled by a DFA and the long-range piece by exact HF exchange (CAM-B3LYP, ωB97X). Essential for charge-transfer excited states (TDDFT) and for asymptotic potentials $v_{\mathrm{xc}} \to -1/r$.

Double hybrids: include a fraction of virtual-orbital (MP2-like) correlation. State-of-the-art accuracy for thermochemistry and barrier heights, but $\mathcal{O}(N^5)$ scaling and very basis-set hungry.

4.3 Self-interaction error

For a one-electron system the electron cannot interact with itself, so $E_{\mathrm{H}} + E_{\mathrm{xc}} = 0$ exactly. In HF theory this cancellation is enforced identity by identity: the diagonal Coulomb J_{ii} is killed by the diagonal exchange K_{ii}. **In approximate KS-DFT this cancellation is broken:**

$$E_{\mathrm{H}}[\rho_i] + E_{\mathrm{x}}^{\mathrm{DFA}}[\rho_i] \neq 0 \qquad \text{(self-interaction error, SIE).} \tag{17}$$

The hydrogen atom is the canonical diagnostic: LDA gives $E(\mathrm{H}) = -0.479$ Ha versus the exact -0.500 Ha; the 0.021 Ha discrepancy is pure SIE in a single-electron system.

Error source: Self-interaction & delocalisation (many-electron SIE)

The many-electron generalisation is the **delocalisation error**: approximate functionals are not piecewise linear in fractional electron number ($\partial E/\partial N$ should be piecewise constant on $(N-1, N)$ and $(N, N+1)$; for LDA/GGA the curve is concave). Practical consequences:

- Stretched bonds dissociate to fractional charges (H_2^+ at large R gives a too-low energy; H_2 at large R a too-high energy).
- Reaction barriers systematically too low ($\sim$ 5 kcal/mol).
- Charge-transfer excited states at wrong energies (often too low).
- Polaron stabilisation too weak; over-delocalised hole states in oxides.
- Spurious self-trapping or non-trapping depending on system.

Controlling the error

(a) Hybrid / range-separated functionals. A fraction a of exact exchange cancels a fraction a of SIE for one-electron densities. Range-separated hybrids enforce the correct $-1/r$ asymptote.

(b) Perdew–Zunger self-interaction correction (PZ-SIC). Subtract the spurious self-Hartree–XC orbital by orbital: $E_{\mathrm{xc}}^{\mathrm{SIC}}[\{\rho_i\}] = E_{\mathrm{xc}}^{\mathrm{DFA}}[\rho] - \sum_i \big(E_{\mathrm{H}}[\rho_i] + E_{\mathrm{xc}}^{\mathrm{DFA}}[\rho_i, 0]\big)$. Computa-

tionally awkward (orbital-dependent potential breaks size-consistency and unitary invariance); production-grade implementations exist (e.g. *Pederson–Lin–Yang* variants).
(c) DFT+U. For localised d or f electrons, add an on-site Hubbard correction (Liechtenstein, Dudarev formulations):

$$E_{\mathrm{DFT}+U} = E_{\mathrm{DFT}} + \frac{U_{\mathrm{eff}}}{2}\sum_{I,\sigma}\mathrm{Tr}\big[\boldsymbol{n}^{I\sigma}(\mathbb{1} - \boldsymbol{n}^{I\sigma})\big], \qquad U_{\mathrm{eff}} = U - J, \tag{18}$$

penalising fractional on-site occupations. The cure for SIE-driven over-delocalisation in 3d transition-metal oxides, lanthanides, actinides. U_{eff} is system-dependent: choose it from linear-response theory (Cococcioni–de Gironcoli) rather than "fitting to gap".
(d) Koopmans-compliant functionals (more recent): enforce piecewise linearity by construction.

4.4 The band-gap problem and the derivative discontinuity

For the *exact* Kohn–Sham potential,

$$\varepsilon^{\mathrm{KS}}_{\mathrm{HOMO}} = -I \quad \text{(ionisation potential)}, \qquad \varepsilon^{\mathrm{KS}}_{\mathrm{LUMO}} - \varepsilon^{\mathrm{KS}}_{\mathrm{HOMO}} \neq E_g^{\mathrm{fund}}. \tag{19}$$

The fundamental gap $E_g^{\mathrm{fund}} = I - A$ (ionisation potential minus electron affinity) differs from the KS gap by the **derivative discontinuity** Δ_{xc}:

$$\boxed{E_g^{\mathrm{fund}} = \left(\varepsilon^{\mathrm{KS}}_{\mathrm{LUMO}} - \varepsilon^{\mathrm{KS}}_{\mathrm{HOMO}}\right) + \Delta_{xc}} \tag{20}$$

Δ_{xc} is the jump in v_{xc} as N crosses an integer. Continuum LDA/GGA functionals have $\Delta_{xc} = 0$ by construction; their KS gap underestimates E_g^{fund} by 30–50% for typical semiconductors.

Error source: Band-gap underestimation

PBE band gap of Si: ~ 0.6 eV; experimental: 1.17 eV. PBE gap of ZnO: ~ 0.8 eV; experimental: 3.4 eV. This is *not* convergence error — it is a structural feature of LDA/GGA v_{xc}.

Controlling the error

(a) Hybrid functionals introduce a non-zero Δ_{xc} *via* the exact-exchange admixture. HSE06 (range-separated, screened HF) is the workhorse for solid-state gaps — gap of Si: 1.15 eV; ZnO: ~ 2.5 eV.
(b) DFT+U can open gaps in strongly-correlated insulators (NiO, FeO).
(c) Many-body perturbation theory (GW). Computes the quasiparticle gap directly. G_0W_0@PBE: ZnO ~ 3.3 eV. Costly; usually performed on top of a converged DFT calculation.
(d) Scissors operator. Empirical: shift all conduction bands rigidly by $(E_g^{\mathrm{exp}} - E_g^{\mathrm{DFT}})$. Cheap, harmless for many post-processing tasks (optics, dielectric function) but *not* a fix — the orbital character is still PBE.

4.5 Static / strong correlation

Single-determinant KS-DFT — like HF — breaks down when the ground state has substantial multi-reference character (stretched bonds, degenerate or near-degenerate frontier orbitals, transition-metal dimers, biradicals).

Error source: Static correlation failures

H_2 at large R, Cr_2, NiO Néel state, low-spin/high-spin energetics, transition states of bond-forming reactions. Symptoms: unstable wave function (triplet–singlet near-degeneracy), strong functional dependence, large spin contamination in UKS.

Controlling the error

(a) Run a stability analysis (ORCA: `! STAB`); if unstable, follow the instability mode.
(b) Use broken-symmetry UKS with $\langle \hat{S}^2 \rangle$ monitoring; project spin where appropriate.
(c) Escalate to multi-reference methods: CASSCF(/NEVPT2), DMRG, MRCI, or *ensemble / state-averaged* DFT.
(d) Specialised functionals (multi-configuration pair-density functional theory, MC-PDFT) for moderate cases.

4.6 Van der Waals / dispersion

Standard LDA, GGA, meta-GGA and hybrid functionals are local or semi-local in the density. They cannot describe the long-range $-C_6/R^6$ attraction between non-overlapping density fragments.

Error source: Missing dispersion

π-stacking (DNA bases, graphene layers), molecular crystals (paracetamol, urea), physisorption energies, layered materials (MoS_2, graphite interlayer spacing), gas-phase dimers (Ar_2, benzene dimer). Bare PBE predicts graphite layers unbound.

Controlling the error

(a) Pairwise atom–atom corrections (Grimme): D2, D3, D3(BJ), D4 — add $E_{\text{disp}} = -\sum_{IJ} f_{\text{damp}}(R_{IJ})\, C_6^{IJ}/R_{IJ}^6 - (C_8 \text{ terms})$. Cheap; default choice for most chemistry. Activated by appending `-D3` to the functional name.
(b) Tkatchenko–Scheffler (TS), MBD: C_6 coefficients computed from the in-situ density via Hirshfeld partitioning; many-body dispersion (MBD) includes beyond-pairwise terms. Essential for highly polarisable systems.
(c) Non-local vdW-DF functionals: vdW-DF, vdW-DF2, rev-vdW-DF2, optB88-vdW — include a non-local correlation kernel directly in $E_c[\rho]$. More expensive but parameter-free.
(d) Rule of thumb: for any system held by non-covalent forces, do *not* run bare PBE; always add D3 (or D4) at minimum.

4.7 Matching XC functionals to experimental scenarios

Deliverable: XC–experiment matching worksheet

For each scenario in the left column, identify a sensible default functional, the principal expected error, and an escalation step if higher accuracy is needed.

Experimental scenario	**Default**	**Principal expected error**	**Escalation**
Geometry / lattice constants of metals	PBE or PBEsol	PBE: $\sim$ 1% overestimate of a; PBEsol corrects this for densely-packed solids	SCAN
Geometry of molecular crystals (paracetamol, energetic materials)	PBE-D3(BJ)	Without D3, lattice volumes too large by 5–15%	TS, MBD; SCAN+rVV10
Bulk modulus / phonon frequencies of metals	PBE (or PBEsol) + Marzari–Vanderbilt cold smearing	Phonon frequencies sensitive to functional and $\boldsymbol{k}$-grid	SCAN
Band gap of a semiconductor or insulator	HSE06	PBE underestimates by 30–50%; HSE06 typically within $\sim$ 0.3 eV of experiment for sp semiconductors	G_0W_0, scQPGW
Band gap of a transition-metal oxide (NiO, MnO)	PBE+U or HSE06	PBE predicts metallic when experiment is insulating (NiO)	DFT+DMFT, GW
IR spectrum of a covalent molecule (FTIR of CO_2, acetone, etc.)	B3LYP or PBE0 / def2-TZVP	Harmonic frequencies $\sim$ 3–5% high; use empirical scaling factor	meta-GGA, anharmonic corrections
Raman spectrum of an oxide (TiO_2 anatase)	PBE / norm-conserving or PAW	Raman activities sensitive to $\boldsymbol{k}$ grid and functional; LO–TO splitting needs Born charges	SCAN, hybrid PBE0
UV–Vis / TDDFT of an organic chromophore	ωB97X-V or CAM-B3LYP	B3LYP underestimates charge-transfer excitations	ADC(2), DLPNO-STEOM-CCSD
Reaction barrier (transition state)	PBE0 or M06-2X	PBE: barriers $\sim$ 5 kcal/mol low (SIE)	double hybrid, DLPNO-CCSD(T)
Adsorption on a metal surface (Pt, Pd)	RPBE / BEEF-vdW	PBE chemisorption energies $\sim$ 0.2–0.5 eV overbound	SCAN+rVV10, RPA
Layered material (graphite, MoS_2, h-BN)	optB88-vdW or rev-vdW-DF2	Interlayer spacing wrong by $>$ 10% without dispersion	MBD
Strongly correlated f-electron actinide oxide (UO_2, PuO_2)	PBE+U (with care in choice of U)	Wrong magnetic ground state without U; sensitivity to U value	DFT+DMFT
Free-radical chemistry, biradicals	BS-UKS (broken-symmetry) + spin projection	Spin contamination; multi-reference character	CASSCF/NEVPT2

What to tell experimentalists

Three-line summary for students.
(1) For most periodic solids and surfaces: PBE (+D3 if any weak interactions).
(2) For molecular thermochemistry and vibrational spectroscopy: B3LYP or PBE0 + def2-TZVP (+D3).
(3) For band gaps of semiconductors and insulators: HSE06.
For everything else, consult §4.7 and *never* report a single-functional result without a sensitivity check against at least one other rung of the ladder.

5 Basis Sets and Pseudopotentials

1:25 – 1:45 (20 min)

Aim: contrast plane-wave (QE) and localised (ORCA) representations; cover pseudopotential / PAW choices; expose basis-set incompleteness, BSSE, frozen-core, and linear-dependence errors and their controls.

5.1 Two families of basis sets

The KS orbitals must be expanded in some finite set $\{\chi_\mu(\boldsymbol{r})\}$ for numerical solution. The two dominant choices in this course:

Plane waves (QE). Periodic Bloch states are expanded as

$$\psi_{n\boldsymbol{k}}(\boldsymbol{r}) = \frac{1}{\sqrt{\Omega}} \sum_{\boldsymbol{G}} c_{n,\boldsymbol{k}+\boldsymbol{G}}\, e^{i(\boldsymbol{k}+\boldsymbol{G})\cdot\boldsymbol{r}}, \tag{21}$$

truncated by the kinetic-energy cutoff

$$\tfrac{1}{2}|\boldsymbol{k}+\boldsymbol{G}|^2 \le E_{\mathrm{cut}}^{\psi} \quad (\texttt{ecutwfc}). \tag{22}$$

A second cutoff E_{cut}^{ρ} = `ecutrho` bounds the plane waves used to represent $\rho(\boldsymbol{r})$ and $v_{\mathrm{eff}}(\boldsymbol{r})$. For norm-conserving PPs: `ecutrho` = 4× `ecutwfc` (exact for the Hartree integral to be unaliased); for ultrasoft / PAW: typically 8–12× `ecutwfc`.

Pros: systematically improvable (single parameter `ecutwfc`); basis is independent of atomic positions, so no Pulay forces; perfectly periodic by construction; FFT-friendly. Cons: enormous basis (10^4–10^6 functions for a single $\boldsymbol{k}$-point) because plane waves represent a node every $\sim 2\pi/|\boldsymbol{G}|_{\max}$ everywhere in space, including in vacuum.

Localised (Gaussian) basis (ORCA). Each atomic orbital is a linear combination of contracted Gaussian primitives times a spherical harmonic:

$$\chi_\mu(\boldsymbol{r}) = Y_\ell^m(\hat{\boldsymbol{r}}) \sum_p d_{\mu p}\, e^{-\alpha_{\mu p}|\boldsymbol{r}-\boldsymbol{R}_A|^2}. \tag{23}$$

Pros: tiny basis (10^1–10^3 functions), efficient analytic integrals (McMurchie–Davidson, Obara–Saika); natural for molecules; hybrid functionals affordable. Cons: not systematically improvable in one knob (one must pick a basis family and a cardinality); basis moves with atoms (Pulay forces); basis-set superposition error (BSSE); periodicity awkward.

5.2 Localised basis families used in ORCA

Karlsruhe def2 series. `def2-SV(P)`, `def2-SVP`, `def2-TZVP`, `def2-TZVPP`, `def2-QZVP`, `def2-QZVPP`. "`SV`/`TZ`/`QZ`" = split-valence / triple-zeta / quadruple-zeta; "`V`" = valence (only the valence is split); "`P`" / "`PP`" = polarisation functions on heavy/all atoms. Diffuse-augmented variants: `def2-SVPD`, `def2-TZVPD`. **Recommended defaults**: `def2-SVP` (cheap survey), `def2-TZVP` (production), `def2-QZVP` (benchmark).

Dunning correlation-consistent. `cc-pVDZ`, `cc-pVTZ`, `cc-pVQZ`, `cc-pV5Z`, with diffuse-augmented `aug-cc-pVXZ`. Designed for systematic convergence of correlation energy and for extrapolation to the complete basis set limit.

Auxiliary basis sets for resolution-of-the-identity (RI). ORCA's `def2/J` (Coulomb fit) and `def2/JK` (Coulomb + exchange fit) accelerate the four-centre integral evaluation by 10–100× with negligible accuracy loss; always use them for hybrid functionals (`! RIJCOSX def2/J`).

5.3 Basis-set error: incompleteness

Error source: Basis-set incompleteness

A finite basis spans a subspace of L^2; the variational energy is always above the true KS minimum. Typical magnitudes:

- `def2-SVP`: total energies $\sim$ 0.05 Ha above CBS; relative energies (atomisation, reaction) $\sim$ 5–10 kcal/mol from CBS.
- `def2-TZVP`: $\sim$ 1–2 kcal/mol from CBS for most reactions.
- `def2-QZVP`: $\sim$ 0.3 kcal/mol from CBS.

Controlling the error

(a) Cardinality ladder. Run the same calculation at DZ, TZ, QZ; report the spread.
(b) Complete-basis-set (CBS) extrapolation. For correlation-consistent bases,

$$E(X) = E_{\text{CBS}} + A\,X^{-3} \qquad (X = 2, 3, 4, \ldots), \tag{24}$$

typically fit on $X = 3, 4$ for HF + correlation separately.
(c) Augmentation with diffuse functions (`aug-` or `def2-...D`) for anions, excited states, weakly-bound complexes, NMR shieldings, polarisabilities.

5.4 Basis-set superposition error (BSSE)

When two fragments A and B form a complex AB, in the supermolecular calculation each fragment can "borrow" the basis functions of the other, lowering its monomer energy and *artificially deepening* the interaction energy.

Error source: BSSE in molecular complexes

Typical magnitudes at `def2-SVP`: 1–3 kcal/mol for hydrogen bonds; smaller but non-negligible at `def2-TZVP`. Crucial in lesson 4 (IR of molecular clusters) and lesson 5 (Raman of host–guest systems).

Controlling the error

Counterpoise correction (Boys–Bernardi): compute each monomer's energy in the full dimer basis ("ghost" atoms with basis but no nucleus or electrons):

$$\Delta E_{AB}^{CP} = E_{AB}(AB_{basis}) - E_A(AB_{basis}) - E_B(AB_{basis}). \tag{25}$$

ORCA syntax: `! GCP(DFT/TZ)` (geometric counterpoise — a fast *a priori* estimate); or run three single-points with ghost atoms. BSSE vanishes by construction at the CBS limit, so the principal cure is a bigger basis.

5.5 Linear dependence and ill-conditioning

Augmenting a localised basis with many diffuse functions makes the overlap matrix $\mathbf{S}$ near-singular: small eigenvalues of $\mathbf{S}$ (below $\sim 10^{-6}$) inflate numerical noise and can cause SCF instability.

Controlling the error

Canonical orthogonalisation: diagonalise $\mathbf{S} = \mathbf{U}\boldsymbol{\sigma}\mathbf{U}^\top$, discard columns of $\mathbf{U}$ with $\sigma_k < \sigma_{thr}$ ($\sigma_{thr} \sim 10^{-7}$ typical), use $\mathbf{X} = \mathbf{U}_{kept}\,\boldsymbol{\sigma}_{kept}^{-1/2}$ as transformer. ORCA: `%scf STOL 1e-7 end`. Implemented in your notebook's SCF routine.

5.6 Pseudopotentials and PAW

Core electrons are chemically inert and have very deep v_{ext} near the nucleus, requiring (in plane waves) enormous E_{cut} to represent. **Pseudopotentials** (PPs) replace the singular nuclear potential and the core electrons with a smooth effective potential whose lowest valence eigenstates match the all-electron ones outside a cutoff radius r_c.

Norm-conserving PPs (NCPP). The pseudo-wavefunction has the same norm inside r_c as the all-electron wavefunction (Hamann–Schlüter–Chiang 1979; Troullier–Martins 1991). Hard (large `ecutwfc`, often ≥ 80 Ry), but compatible with virtually all post-DFT methods.

Ultrasoft PPs (USPP). Vanderbilt (1990) relaxed norm-conservation, requiring an augmentation charge to restore the correct density. Much softer (`ecutwfc` ~ 30 Ry typical) but requires a denser charge-density grid (`ecutrho` $\approx$ 8–10$\times$`ecutwfc`).

Projector-augmented-wave (PAW). Blöchl (1994): an exact transformation between all-electron and pseudo-wavefunctions, expressed via local projectors. The PAW method recovers all-electron quantities (densities, magnetic moments, NMR, EFG) at near-USPP cost. *The recommended default for most modern QE calculations.*

Error source: Pseudopotential / PAW error

- *Frozen-core error*: PPs assume the core is unchanged by the chemical environment. For 3d transition metals (Sc–Zn) and lanthanides, the semicore $3s, 3p$ ($4s, 4p$, $5s, 5p$) shells should be *included in the valence*. Use "`_3s3p`" or "semicore" variants of the PP.
- *r_c choice*: if r_c is too large, the PP is too soft and the chemistry around the atom is distorted ("ghost states", wrong bond lengths). If too small, E_{cut} must be huge.
- *Transferability*: a PP fitted to the isolated atom may underperform in a different bonding environment (charged ions, high pressure).
- *XC consistency*: a PP generated with PBE *should* be used with PBE. Cross-using (PBE PP + LDA calculation) introduces uncontrolled errors.

Controlling the error

(a) Use a curated library. The **Standard Solid-State Pseudopotentials** (SSSP, Prandini et al., *npj Comput. Mater.* 2018, maintained at materialscloud.org) provides "efficiency" and "precision" sets benchmarked against all-electron WIEN2k for ~ 70 elements. The SSSP-precision set is the recommended default for new QE calculations; SSSP-efficiency for high-throughput screening.
(b) Match XC. Use a PP generated with the same XC as the calculation.
(c) Use semicore PPs for 3d transition metals, lanthanides, actinides, and for high-pressure or high-charge-state work.
(d) ORCA equivalent. For heavy elements ($Z \geq 36$), use def2-ECP (Stuttgart effective core potentials) bundled with the def2 basis. Light elements: all-electron, no ECP needed.
(e) Cutoff convergence. For every new PP/PAW, run an `ecutwfc` sweep on a representative system (typically the bulk crystal) and converge total energies to within 1 meV/atom; `ecutrho` to similar tolerance.

6 Periodic Boundary Conditions and $\boldsymbol{k}$-point Sampling

1:45 – 2:00 (15 min)

Aim: introduce Bloch's theorem, the Brillouin zone, Monkhorst–Pack $\boldsymbol{k}$-grids, smearing for metals, supercell convergence for defects and isolated molecules in PBC.

6.1 Bloch's theorem

In a perfect crystal with Bravais lattice $\{\boldsymbol{R}\}$ and corresponding reciprocal lattice $\{\boldsymbol{G}\}$, the effective potential is periodic: $v_{\text{eff}}(\boldsymbol{r} + \boldsymbol{R}) = v_{\text{eff}}(\boldsymbol{r})$. The KS Hamiltonian commutes with all lattice translations, so eigenstates are simultaneous eigenstates of the translation operators. Bloch's theorem:

$$\psi_{n\boldsymbol{k}}(\boldsymbol{r}) = e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\, u_{n\boldsymbol{k}}(\boldsymbol{r}), \qquad u_{n\boldsymbol{k}}(\boldsymbol{r} + \boldsymbol{R}) = u_{n\boldsymbol{k}}(\boldsymbol{r}), \tag{26}$$

with band index n and crystal momentum $\boldsymbol{k}$ in the first Brillouin zone (BZ). The KS problem is therefore solved *separately at each* $\boldsymbol{k}$, then averaged.

6.2 Brillouin-zone integration

Any extensive quantity is an integral over the BZ. Total energy per cell:

$$E_{\text{tot}} = \frac{1}{V_{\text{BZ}}} \sum_n \int_{\text{BZ}} f_{n\boldsymbol{k}}\, \varepsilon_{n\boldsymbol{k}}\, \mathrm{d}\boldsymbol{k} + (\text{double-counting corrections}), \tag{27}$$

discretised on a finite mesh $\{\boldsymbol{k}_j\}_{j=1}^{N_{\boldsymbol{k}}}$ with weights w_j:

$$\int_{\text{BZ}} g(\boldsymbol{k})\, \mathrm{d}\boldsymbol{k} \;\longrightarrow\; \sum_j w_j\, g(\boldsymbol{k}_j). \tag{28}$$

Monkhorst–Pack grid. The standard prescription: a uniform mesh of size $n_1 \times n_2 \times n_3$ in the reduced reciprocal coordinates, optionally shifted by $(s_1/2, s_2/2, s_3/2)$. Time-reversal symmetry and crystal symmetry are exploited to reduce to the *irreducible* BZ (IBZ).

$\boldsymbol{k}$-grid heuristics. Use roughly equal density of $\boldsymbol{k}$-points along each reciprocal direction: $n_i \propto 1/|\boldsymbol{a}_i|$. Convenient figure of merit: $\boldsymbol{k}$-point *spacing* $\Delta k = 2\pi/(n_i|\boldsymbol{a}_i|)$. Typical converged values:

- Insulators / semiconductors: $\Delta k \lesssim 0.03$ Å^{-1} ($\sim 8 \times 8 \times 8$ for $a \sim 5$ Å).

- Metals: $\Delta k \lesssim 0.02$ Å^{-1} (denser is needed because the Fermi-surface integrand is discontinuous).
- Large supercells or molecules in a box: $\boldsymbol{k}$-grid becomes redundant; Γ-point only is then exact.

Error source: $\boldsymbol{k}$-point sampling error

Coarse $\boldsymbol{k}$-grids give absolute energies that differ from the converged value by hundreds of meV. Total energies, lattice constants, bulk moduli, phonon frequencies, magnetic moments, all depend on the $\boldsymbol{k}$-grid. For metals the dependence is non-monotonic and can be deceptively smooth at the wrong value.

Controlling the error

Always perform a $\boldsymbol{k}$-convergence test on a representative system: e.g. $4^3, 6^3, 8^3, 10^3, 12^3$ at fixed `ecutwfc` and `degauss`. Plot $E_{\text{tot}}(N_{\boldsymbol{k}})$. Choose the smallest grid for which energy is converged to your target tolerance (typically 1 meV/atom). For metals, *simultaneously* vary smearing σ (§3.5.2).

6.3 The Γ-point approximation

For supercells large enough that the BZ becomes small ($\boldsymbol{a}_i \gtrsim 10$ Å), a single $\boldsymbol{k}$-point at $\Gamma = (0, 0, 0)$ suffices. Use cases:

- Molecules in a large vacuum box (the "periodic-image molecule" trick to use a plane-wave code on a molecule);
- Point defects in large supercells (e.g. vacancy in $4 \times 4 \times 4$ Si);
- Disordered systems (amorphous Si, liquid water) in ≥ 64-atom cells.

Error source: Spurious periodic-image interactions for molecules in PBC

A charged or polar molecule in a finite cell electrostatically interacts with its periodic images. Energy errors scale as q^2/L (monopole), p^2/L^3 (dipole), and so on.

Controlling the error

(a) Use a large enough box (typically ≥ 15 Å of vacuum between images for neutral molecules; ≥ 20 Å for dipolar; *much* more for charged).
(b) Apply electrostatic decoupling: **Makov–Payne** correction for charged cells (cubic only); **Martyna–Tuckerman** reciprocal-space decoupling (general); QE keyword `assume_isolated = 'mt'` (Martyna–Tuckerman) or `'mp'` (Makov–Payne).
(c) For molecular spectroscopy in QE, ORCA is usually a cleaner choice unless periodicity is essential.

6.4 Practical $\boldsymbol{k}$-paths for band structures (briefly — detailed in Lesson 3)

Band structures are computed along high-symmetry $\boldsymbol{k}$-paths in the IBZ (e.g. Γ–X–M–Γ–R for simple cubic; Γ–K–M–Γ–A for hexagonal). Tools: `seekpath` (Hinuma et al., 2017) and the AFLOW conventions provide canonical paths. The `bands.x` post-processor in QE reads a non-SCF run along such a path.

7 Closing summary, validation protocol, references

Final 5 min

Synthesise: every KS-DFT calculation has *five* sources of error; every published result should declare control over each.

7.1 The five-axis error map

Axis	Origin	Control / convergence parameter
1. Exchange–correlation	Unknown exact $E_{xc}[\rho]$ approximated by functional	Functional class (LDA $\to$ GGA $\to$ meta-GGA $\to$ hybrid); cross-check with different rungs
2. Self-interaction	DFA self-Hartree not cancelled by self-exchange	Hybrid / range-separated; PZ-SIC; DFT+U for localised states
3. Basis / cutoff	Finite-dimensional representation of ψ and ρ	`ecutwfc`/`ecutrho` (QE) or cardinality (DZ $\to$ TZ $\to$ QZ); CBS extrapolation; BSSE counterpoise
4. Pseudopotential / PAW	Frozen core, r_c choice, XC inconsistency	SSSP-precision; semicore PPs for transition metals; XC-matched PP; cutoff convergence
5. Brillouin-zone sampling	Discrete $\boldsymbol{k}$-grid, smearing entropy	Monkhorst–Pack convergence; cold smearing σ convergence; image-decoupling for molecules

7.2 Minimum validation protocol for a new calculation

1. **Cutoff convergence.** `ecutwfc` sweep ($\Delta E < 1$ meV/atom).
2. **$\boldsymbol{k}$-grid convergence.** Monkhorst–Pack sweep ($\Delta E < 1$ meV/atom).
3. **Smearing convergence** (metals only). Cold smearing $\sigma = 0.02, 0.01, 0.005$ Ry.
4. **SCF tolerance check.** Tighten `conv_thr` by 10×; result must be invariant within the property tolerance.
5. **Geometry convergence.** `forc_conv_thr` $\leq 10^{-4}$ Ry/Bohr for vibrations.
6. **Functional sensitivity.** Re-run the property with *at least one other functional* (e.g. PBE and SCAN; or PBE0 and B3LYP). Report the spread.
7. **Pseudopotential / basis check.** For at least one critical case, repeat with a higher-cardinality basis or a semicore PP.
8. **Compare with experiment** (or with higher-level theory) on a closely-related, well-characterised reference system.

What to tell experimentalists

Honest reporting. Any DFT result without (1)–(2) is not a result; it is an opinion. With (1)–(4) it is a calculation. With (1)–(7) it is publication-grade. With (8) added it earns the confidence of an experimentalist reading the paper.

7.3 Selected references for the curious

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- G. Prandini *et al.*, *Precision and efficiency in solid-state pseudopotential calculations*, npj Comput. Mater. **4**, 72 (2018) — SSSP library.
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End of Lesson 2. Lesson 3 picks up the SCF cycle as a black box and uses QE / ORCA to actually run band structures, DOS, and geometry optimisations.